E-commerce Customer Behavior and Purchase Intention Analysis

Python • Exploratory Data Analysis • Customer Insights

1. Project Overview

This project focuses on analysing online shopper browsing behaviour and purchase intention using the Online Shoppers Purchasing Intention dataset. It examines session-level activity such as the number of pages viewed, time spent on page groups, bounce and exit rates, page values, visitor type, month, weekend status, and whether a session resulted in revenue.

Python will be used for data inspection, preprocessing, exploratory data analysis (EDA), statistical comparisons and visualization. The goal is to identify patterns associated with purchasing sessions and communicate evidence-based findings and recommendations.

2. Data Source

Dataset Name	Online Shoppers Purchasing Intention Dataset
Source	UCI Machine Learning Repository
Official URL	https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset
File Name	online_shoppers_intention.csv
File Type	CSV
Number of Records	12,330
Number of Attributes	18 original columns
Missing Values	0 reported missing values in the uploaded CSV
Duplicate Rows	125 exact duplicate rows flagged for investigation

The dataset contains browsing-session features and a Boolean Revenue outcome. It does not contain a full transaction date or a customer identifier; Month is provided as a categorical field. The uploaded file includes records for February, March, and May through December, so month-based findings should not be described as complete-year coverage.

3. Problem Statement

E-commerce businesses need to understand how visitor browsing patterns relate to purchase outcomes. Raw session data can be difficult to interpret without systematic analysis, particularly when comparing visitor types, engagement measures, browsing duration and shopping periods.

This project aims to explore the relationships between session behaviour and the Revenue outcome, identify meaningful differences between purchasing and non-purchasing sessions, and present the results through clear visualizations. The analysis will describe associations in the dataset and will not assume that a relationship proves causation.

4. Objectives

- Inspect the structure, data types, value ranges and quality of the uploaded dataset.
- Analyse page views and time spent on administrative, informational and product-related pages.
- Compare browsing and engagement metrics between sessions with Revenue = True and Revenue = False.

- Explore purchase outcomes across visitor types, months, weekend status and traffic-related attributes.
- Examine bounce rates, exit rates and page values in relation to session outcomes.
- Create clear visualizations and summarize findings supported by the data.
- Develop practical, evidence-based recommendations for further e-commerce investigation.

5. Attribute (Column / Feature) Details

Attribute Name	Description
Administrative	Number of administrative pages visited during the session
Administrative_Duration	Time spent on administrative pages
Informational	Number of informational pages visited
Informational_Duration	Time spent on informational pages
ProductRelated	Number of product-related pages visited
ProductRelated_Duration	Time spent on product-related pages
BounceRates	Bounce rate associated with the session/page activity
ExitRates	Exit rate associated with the session/page activity
PageValues	Page-value metric recorded for the session
SpecialDay	Closeness of the browsing date to a special day, as encoded in the dataset
Month	Month label associated with the session
OperatingSystems	Encoded operating-system category
Browser	Encoded browser category
Region	Encoded geographic region
TrafficType	Encoded traffic-source category
VisitorType	Visitor classification, such as returning or new visitor
Weekend	Whether the session occurred on a weekend
Revenue	Whether the session resulted in revenue (target/outcome field)

6. Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook / Google Colab

7. Data Pre-processing and Feature Engineering

The dataset will be checked and prepared before drawing conclusions. Cleaning decisions will be documented so that valid observations are not removed without justification.

7.1 Data Pre-processing

1. Data Loading: Load online_shoppers_intention.csv using Pandas.
2. Data Inspection: Review the number of rows and columns, column names, data types and sample records.
3. Missing Value Analysis: Confirm missing-value counts and inspect any blank or unexpected values.
4. Duplicate Checking: Investigate the 125 exact duplicate rows; remove them only after assessing whether they are repeated records or legitimate repeated sessions.
5. Data Type Validation: Check numeric, Boolean and categorical fields and convert them where appropriate.
6. Range and Consistency Checks: Validate duration, rate and page-value fields for unexpected or invalid values.

7. Category Review: Inspect labels in Month and VisitorType and review encoded fields such as Browser, Region and TrafficType.
8. Outcome Validation: Confirm the meaning and distribution of Revenue = True and Revenue = False.
9. Clean Dataset Preparation: Keep the original CSV unchanged and save a separate cleaned copy if cleaning is performed.

7.2 Feature Engineering

Derived features may be created to support analysis. These are additional analytical features, not original dataset columns.

Feature	Definition	Purpose
Total Pages Viewed	Sum of Administrative, Informational and ProductRelated	Compare overall browsing activity
Total Page Duration	Sum of the three duration fields	Compare time spent browsing
Average Duration per Page	Total page duration divided by total pages viewed when the denominator is greater than zero	Examine browsing depth and time
Purchase Outcome Label	Readable label derived from Revenue	Present conversion outcomes clearly
Month Order	Map Month labels to calendar order for charting	Keep monthly charts chronological

8. Planned Exploratory Analysis and Visualizations

- Distribution of Revenue outcomes and overall purchase-session rate.
- Revenue outcome by VisitorType.
- Revenue outcome by Weekend status.
- Monthly session counts and purchase-session rates for months present in the data.
- Comparison of page views and browsing duration for purchasing versus non-purchasing sessions.
- BounceRates and ExitRates distributions by Revenue outcome.
- PageValues comparison by Revenue outcome.
- Correlation heatmap for selected numeric variables, interpreted cautiously.
- Purchase-session rates by traffic type, browser, operating system and region.
- Relationship between product-related page activity and Revenue outcome.

9. Expected Outcomes

The project is expected to provide a structured view of online shopping sessions, show how browsing and engagement metrics differ across Revenue outcomes, and highlight visitor groups or session patterns that merit further investigation. Actual findings and recommendations will be written only after the analysis is completed.

10. Conclusion

This project uses the Online Shoppers Purchasing Intention dataset to study e-commerce browsing behaviour and purchase outcomes. Python libraries such as Pandas, NumPy, Matplotlib and Seaborn will support data inspection, preprocessing, exploratory analysis and visualization. The work will focus on reliable data-quality checks, comparisons of purchasing and non-purchasing sessions, and clear communication of findings. The final conclusions will be based on observed results and will distinguish association from causation.